DESIGN AND EVALUATION OF AN AUGMENTED REALITY STORYBOOK FOR PRIMARY LEARNERS' PLANT GROWTH UNDERSTANDING & STEWARDSHIP.

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Understanding plant growth and environmental change can be difficult for young children when learning is limited to textbook pictures and verbal explanation. At primary level, children need more interactive and meaningful ways to learn biological processes such as plant health, stress, and recovery. Although Augmented Reality (AR) has improved educational engagement, many existing AR-based plant learning systems mainly focus on showing plant parts or growth stages rather than helping children emotionally interpret plant conditions. As a result, there is still a gap in developing child-friendly AR visualizations that translate plant biological states into simple, meaningful, and age-appropriate emotional cues. This research proposes an Affective Visualization Model for a handheld AR interactive storybook designed for young learners. The model represents plant conditions through AR-optimized 3D assets, animation, color changes, glow effects, motion patterns, and supportive audio cues to communicate health, stress, and recovery. The system will be developed using Unity 3D, AR Foundation, ARCore, ARKit, Blender, and Autodesk Maya. The expected outcomes include improved understanding of plant condition changes, stronger emotional engagement with ecological concepts, and more effective communication of biological responses. Socially, the research supports child-centered environmental education, while commercially it has potential for use in future AR learning products for schools and guided home learning.

Keywords: Augmented Reality, Affective Visualization, Plant Education, Emotion-Responsive Design, Child-Centered Learning

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AR	Augmented Reality
ARCore	Augmented Reality Core
ARKit	Augmented Reality Toolkit
DSR	Design Science Research
UI	User Interface
3D	Three-Dimensional
HCI	Human–Computer Interaction

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Appendix 11.1 Survey Instruments

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1. INTRODUCTION

1.1. Background and Context

Understanding of the process of plant growth and the processes of the environment is a significant part of the of educating young children. However, the biological concepts of plant life, the stages of plant growth, and the effects of environmental stress on plant life are often hard to grasp for young children using conventional teaching methodologies, textbooks, and explanations. This is because these methodologies rarely teach the dynamic response of plants to environmental factors, and as a result, young children often find it hard to grasp the concepts of plant life and its interaction with the surrounding environment. This problem often affects young children, teachers, and educational institutions.

It has been recognized at the international level that environmental education is one of the most important factors that can contribute to the development of societies in a sustainable manner and the promotion of responsible behavior towards nature. Reports from UNESCO on environmental literacy indicate that a large percentage of young learners around the globe are not aware of ecological systems and ideas of sustainability. While there is a need for better educational tools, it has been observed that many existing tools still use diagrams and textual information that cannot effectively show biological processes [1]. While some educational tools use 3D models and animation to show the structures of plants, these tools mainly show the structures of plants and do not show dynamic biological conditions such as the health of the plant. Moreover, emotional design elements are not included in these tools, which can help children intuitively understand the condition of the plant.

The recent trends and developments related to Augmented Reality (AR) technology are providing new opportunities to enhance learning experiences related to science by adding interactive digital content to the real world. AR technology has started to gain popularity for educational purposes because it improves visualization and allows learners to interact with complex ideas in a more engaging way. Research on educational technology has

indicated that AR-based learning environments improve learner engagement, motivation, and learning outcomes to a higher extent compared to traditional learning approaches. [2]. Research has indicated that the use of immersive visualization technologies can improve engagement levels by 30-40%, particularly when complex scientific ideas are presented to learners as interactive visual environments [3]. However, most AR learning tools are limited to showing static 3D objects rather than translating biological states into emotionally understandable forms. Therefore, children may be able to view the structure of plants but still not be able to understand the way plants react to changes in the environment or the way the health conditions of plants evolve.

To address these limits, this research proposes a handheld Augmented Reality (AR) interactive storybook framework to cater to international school children in Sri Lanka aged 6 years and studying Grade 1 Science according to the Cambridge curriculum. The system will be used to enable children to learn about plant growth and environmental factors in an immersive and interactive manner. The overall framework will comprise several research components that will interact with each other to simulate plant growth in a given environment. The components will include a hybrid interaction framework that will link physical storybook pages with AR content, a context-aware simulation framework that will map environmental factors to plant growth parameters, and a narrative framework that will utilize storytelling to enable children to learn about environmental factors in an interactive manner. In this regard, the Affective Visualization Model, which is the main concern of the present research, attempts to transform biological plant states into emotional cues that can be easily understood by children. This includes the development of AR-optimized plant assets, such as 3D plant models of different stages of growth, animations to express the states of the plants, and the exploration of various effects, such as color changes, glow effects, movement patterns, and sound effects. The combination of the aging of the plants, the microscopic features, and the empathetic features is regarded as a successful way to enable children to intuitively understand the states of the plants and empathize with the plants. The research supports United Nations Sustainable Development Goal (SDG) 4 – Quality Education and SDG 15 – Life on Land [4], while aligning with research areas in Augmented Reality, interactive media, and educational visualization.

1.2. Literature Review

1.2.1. Augmented Reality in Science Education

Augmented Reality (AR) technology has been identified as an effective technology in improving learning experiences in science education. A systematic review on the use of AR technology in learning, conducted by Chang, Wu, and Hsu, identified that AR technology can enhance learning engagement, spatial understanding, and learning motivation for learners in comparison to traditional learning approaches [5]. Several researchers have attempted to use AR-based learning systems for interactive learning in science education. A systematic review by Lampropoulos, Keramopoulos, and Diamantaras emphasized the use of AR technology for learning purposes by providing an understanding of complex scientific phenomena [6]. These tools enable abstract ideas to be represented by interactive 3D images. This improves conceptual understanding. However, despite these benefits, existing AR educational tools are still largely focused on providing children with visual information rather than enabling cognitive understanding of the scientific process. Secondly, existing AR tools are still facing usability concerns, such as cognitive overload and complexity, particularly for children [6]. Therefore, although AR technology has been effective for improving engagement, existing tools are still lacking when it comes to providing children with a pedagogically effective way of visualizing complex biological processes.

1.2.2. Augmented Reality in Environmental and Ecological Learning

Environmental and ecological education has also been positively impacted by AR-based learning environments, which allow learners to interact with environmental systems. According to Ladykova, in her systematic review on the impact of augmented reality in environmental education, it has been established that augmented reality applications can improve environmental awareness among students by allowing them to observe ecological systems in virtual environments [7]. In the study titled "Can environmental education supported by augmented reality affect students' environmental knowledge, attitudes, and behaviors?" conducted by Öztürk, it has been established that environmental education supported by augmented reality can improve students' knowledge and attitudes about the environment [8]. Inquiry-based AR learning approaches have also been proposed to

promote the active exploration of ecological concepts and sustainability-related topics, as in the sustainability-based AR learning research by Lo, Lai, and Hsu [9]. However, most of these approaches are based on the dissemination of information on the environment, without visualizing the biological processes that occur in nature. Most of the available environmental-based AR learning approaches present the ecosystem at the macro level, such as forests and habitats, without providing learners with valuable insights on the biological states of the plants, thus enabling them to understand how the environment affects the growth, health, and survival of the plants.

1.2.3. Augmented Reality for Plant-Related Learning

More specialized research has been conducted on AR-based applications for the purpose of supporting plant-related science learning. The AR-based plant experiment system proposed by Jhuang, Lin, and Lin proves that an interactive plant visualization system can improve learning efficiency and reduce the time needed for conducting traditional plant experiments [10]. Such a system allows children to observe the structure of plants with AR-based overlays. This improves accessibility to scientific content, which would normally require long periods of time for conducting experiments. Similarly, AR-based inquiry learning environments for science learning, as proposed by Wen, Chen, and Chen, have been proven to be effective for improving children's motivation and conceptual understanding of natural science content [11]. However, these systems are mainly designed to focus on providing structural visualization for plant anatomy rather than simulating plant states like stress, dehydration, etc. In addition, these plant visualization systems are not designed to include interactive or emotional feedback mechanisms to assist children in understanding plant states. Therefore, although AR technology has improved the accessibility of learning content related to plants, plant visualization systems are not effective in providing plant states with dynamic visualization that can be easily comprehended by children.

1.2.4. Affective and Emotional Design in AR Learning Environments

Recent research has also attempted to investigate the use of emotional and affective design in facilitating user engagement in digital learning environments. A study by Paredes-Velasco, Velázquez-Iturbide, and Gómez-Ríos on the use of AR animation in facilitating user engagement in an AR animation-based learning system revealed that the incorporation of animation in AR-based learning systems can have a significant impact on the emotional state of the user, facilitating motivation in the process [3]. Another study by Aziz, Ahmad, Binsaleh, and Wan A. Rahman on an affective AR-based learning system for visually impaired children also revealed that the incorporation of affective interaction in an AR-based learning system, such as visual cues and audio feedback, can have a significant impact on user engagement in the process [12]. Even though the study revealed that affective interaction can have a significant impact on user engagement, the application of affective interaction in an AR-based ecological and biological learning system has not been explored.

In addition to AR-specific research, emotional design principles for multimedia learning have already confirmed that visual elements such as color, shape, and design are critical for learner motivation and understanding. In the study "Emotional Design in Multimedia Learning: Effects of Shape and Color on Affect and Learning," Plass, Heidig, Homer, and Hayward prove that emotionally designed visual elements can be highly effective for learner motivation and understanding within a digital learning context [13]. Emotionally designed visual interfaces can be highly effective for learner understanding since they can provide a positive emotional experience for the learner. However, most research on emotional visualization has focused on traditional multimedia interfaces rather than AR interfaces. Emotional design principles are rarely combined with biological simulation based on real-world environmental processes. Therefore, there is a need for an integrated system that combines AR interaction, plant biological simulation, and emotional visualization. This is critical for the development of learning technologies that not only visualize plant structure but also communicate plant health, stress, and recovery to children.

1.3. Research Gap

Overall research problem:

The overall research problem of this study is that young children often find it difficult to understand plant growth, environmental changes, and ecological concepts through traditional teaching methods such as textbooks and verbal explanations alone. Although Augmented Reality has improved interactive learning, existing educational AR systems still do not fully support child-centered, emotionally engaging, and biologically meaningful learning experiences for early primary learners.

Component specific research problem:

The research problem of this component specifically focuses on the limited ability of existing AR educational applications to represent plant biological states through emotion-responsive visual cues. While many systems can display plant structures or growth stages, they often fail to show health, stress, and recovery in a way that young children can easily interpret. As a result, children may see the content visually but not emotionally understand plant conditions.

While Augmented Reality technologies have been employed in educational environments, previous research in AR in educational settings primarily aims to enhance visualization and learner engagement, rather than enabling the translation of scientific content into an emotionally interpretable experience. Previous studies indicate that learners benefit in terms of motivation, spatial understanding, and learning outcomes through immersive visualization and interactive experiences [5], [6]. Similarly, AR-based educational applications in environmental education have been found to enhance ecological awareness and learners' understanding of sustainability concepts [7], [8]. However, most AR-based educational applications in ecological learning mainly allow learners to visualize information and do not translate ecological conditions and sustainability concepts into emotionally interpretable experiences. As a result, learners are able to understand

ecological content at a surface level, but they are not effectively supported in relating these concepts to real-world biological responses. This indicates the need for an AR system that translates ecological and sustainability concepts into emotionally interpretable experiences through affective visualization technologies.

Identified Research Gap 1 - Limited use of affective visualization in ecological AR learning:

Existing AR educational systems mainly focus on improving visualization and engagement, but they rarely translate ecological and plant-related concepts into emotionally interpretable experiences for young children. As a result, learners may observe content through AR, yet they are not effectively guided to emotionally understand plant conditions and biological responses. [5], [8]

Another limitation identified in existing AR-based learning for plants is the inability of most systems to represent dynamic biological states. For example, the AR plant experiment system proposed by Jhuang, Lin, and Lin allows students to access learning related to plant anatomy and the experimental process through AR technology [10]. Similarly, inquiry-based learning for plants using AR has been found to enhance students' motivation and conceptual understanding of science-related learning [11]. However, although these approaches improve accessibility to plant-related learning, they are limited in representing the biological responses of plants to environmental conditions such as dehydration or stress. Therefore, although a learner can access learning related to plant structures, he or she may still be unable to understand the effect of environmental conditions on the actual state of plants.

Identified Research Gap 2 - Inadequate representation of dynamic plant biological states:

Many AR-based plant learning applications support the visualization of plant anatomy, growth stages, or science experiments, but they do not sufficiently represent dynamic biological conditions such as health, stress, dehydration, and recovery. This limits children's ability to understand how environmental factors affect living plants over time. [10], [11]

In addition, there are limitations in AR learning systems that do not incorporate child-oriented affective design principles. Research has been conducted to understand emotional interaction in AR learning environments, showing that animation can play an important role in learning [3]. Affective AR learning environments designed for inclusive education have also demonstrated that multimodal feedback can contribute positively to children's learning experiences [12]. However, affective AR learning environments have not been adequately developed for ecological learning contexts. Most existing affective AR learning environments focus on mathematics learning, algorithm learning, or accessibility learning. Therefore, emotional visualization techniques are still not commonly used in ecological learning environments. Since children require intuitive visual cues to understand abstract biological concepts, the lack of child-oriented affective visualization mechanisms remains an important gap in the development of AR learning environments.

Identified Research Gap 3 - Lack of child-oriented affective design mechanisms in plant learning environments:

Although affective AR learning environments and multimodal feedback approaches have been explored in other educational domains, such as mathematics, accessibility, and algorithm learning, similar child-centered affective visualization approaches have not been adequately applied to ecological or plant-based AR education. [3], [12]

Finally, although research on AR interaction, environmental education, and emotional design within multimedia systems has been individually studied, there are no strongly integrated works that combine these elements within a single framework. Emotional design principles show that color, shape, and visual appeal can affect learner motivation and understanding within a digital learning system [13]. At the same time, AR-based educational systems mainly focus on AR visualization and interaction, while environmental education research mainly concentrates on knowledge acquisition and awareness [5], [7]. However, these elements are usually studied separately rather than being integrated into a unified educational solution. Therefore, there are still limited integrated solutions for AR interaction, plant biological simulation, and affective visualization. This issue should be addressed in order to develop educational technology that not only allows children to learn plant structures, but also helps them understand plant health, stress, and recovery in a meaningful and emotionally engaging way.

Identified Research Gap 4 - Limited integration of AR interaction, plant simulation, and emotional design within a single framework:

Prior studies have investigated AR interaction, environmental education, and emotional design separately, but there is still a lack of integrated systems that combine these elements into one educational solution. This creates a gap in the development of learning environments that help children understand both plant structures and plant condition changes in a meaningful and emotionally engaging way. [5], [7], [13]

Table 1 : Research Gap Comparison

Gap	What Existing Studies Address	What is Still Missing	Proposed Contribution
Research Gap 1: Limited use of affective visualization in ecological AR learning	Existing AR studies improve visualization, engagement, and environmental understanding in learning contexts. [5], [7], [8]	Emotional interpretation of ecological and plant-related concepts for young children is still limited. [5], [8]	The proposed research introduces affective visualization to make plant conditions more emotionally understandable for children.
Research Gap 2: Inadequate representation of dynamic plant biological states	Existing plant-based AR systems support plant structure, anatomy, and inquiry learning. [10], [11]	Dynamic states such as health, stress, dehydration, and recovery are not clearly represented. [10], [11]	The proposed system visualizes changing plant states through AR-based dynamic representations.
Research Gap 3: Lack of child-oriented affective design mechanisms in plant learning environments	Affective AR studies show that animation and multimodal feedback can improve user engagement. [3], [12]	Similar child-centered affective approaches are not adequately applied to ecological or plant-based AR learning. [3], [12]	The research applies child-oriented affective design using color, glow, motion, animation, and audio cues.
Research Gap 4: Limited integration of AR interaction, plant simulation, and emotional design within a single framework	Previous studies explore AR interaction, environmental education, and emotional design. [5], [7], [13]	There is no strong integrated framework combining these elements for plant learning. [5], [7], [13]	The proposed research combines AR interaction, plant simulation, and affective visualization in one learning framework.

2. OBJECTIVE

2.1. Main Objective

To design, develop, and evaluate a child-centered handheld Augmented Reality (AR) interactive storybook application that supports Sri Lankan international school children aged 6 studying the Cambridge Grade 1 Science curriculum in understanding plant growth, environmental interactions, and ecological responsibility through immersive, intuitive, and age-appropriate learning experiences. The proposed system aims to combine physical storybook interaction with AR visualization, affective plant representation, gamified storytelling, and context-aware environmental simulation to create a meaningful educational environment for young learners. By integrating these elements into a single framework, the research seeks to move beyond conventional static learning methods and provide a more engaging, interactive, and emotionally supportive approach to science education. The overall objective of the study is to improve children's conceptual understanding of plant-related science concepts while also encouraging curiosity, empathy toward living things, and stronger engagement with environmental learning through an accessible handheld AR solution.

2.2. Specific Objectives

The specific objectives of this research are derived from the main objective and are intended to guide the design, development, and evaluation of the proposed affective visualization model in a structured manner. These objectives focus on identifying suitable visual communication methods for young learners, developing AR-ready plant assets, implementing animation and affective visual effects to represent plant biological states, and integrating the model into the overall AR interactive storybook system. Together, these objectives ensure that the research addresses both the educational needs of the target age group and the technical requirements of creating a child-centered AR learning experience.

The following specific objectives have been identified:

1. To analyze the Cambridge Grade 1 Science curriculum and children's visual perception preferences to identify suitable visual cues such as color, motion, animation, and audio that effectively communicate plant growth and environmental concepts to six-year-old learners.
2. To design and develop AR-optimized 3D plant models representing different plant growth stages using 3D modeling software such as Autodesk Maya and Blender for integration into the AR learning application.
3. To create and implement animation sequences and affective visualization techniques, including color transitions, glow effects, motion patterns, and audio cues, to represent plant biological states such as health, stress, and recovery in a way that enhances emotional readability for young learners.
4. To integrate the affective visualization model into the Unity-based AR interactive storybook application and evaluate its effectiveness in helping children understand plant biological states and environmental interactions.

3. METHODOLOGY

This research uses the Design Science Research (DSR) approach, which emphasizes the designing and evaluation of technological solutions that solve real-world problems. The technological solution in this research is an Affective Visualization Model that will be part of an Augmented Reality (AR) educational application for children. The reason for using DSR in this research is that it enables the design, development, and evaluation of innovative technological solutions that are both technological and educational in nature. The previous research on Augmented Reality-based educational environments found that interactive visualization technologies could enhance the learning experience of children in science education [5], [6]. The research will be carried out based on the design of science research methodology. The steps involved in the research are identifying the problem, designing the solution, developing the system, and evaluating the system. The system will be part of an AR interactive storybook for children in international schools in Sri Lanka. The children are aged 6 years and are studying the Cambridge Grade 1 Science curriculum.

The technical methodology for this research involves the development of an affective visualization model that maps biological plant conditions to emotional visual feedback. The system will be able to visually represent various plant conditions such as healthy growth, environmental stress, and recovery by incorporating various visual feedback mechanisms such as color changes, animation behaviors, and glow effects. These visualization mechanisms are based on previous research on animation and visual feedback within an AR environment, which has a positive effect on emotional engagement and learning motivation for children [12]. The plant visualization module will include a procedural model for plant growth and various stages of plant aging. Changes to plant conditions will trigger corresponding visual feedback mechanisms. Environmental conditions such as light and movement will affect plant growth conditions through the context-aware simulation module. Previous research on AR-based learning environments for children to learn environmental conditions emphasized the need for interactive simulation to enhance children's understanding of ecological conditions [7], [9]. The proposed research methodology will be able to provide a more intuitive understanding of plant health conditions by incorporating biological simulation with affective visualization.

The development environment will be created with the help of Unity, Autodesk Maya, and Blender, depending on their suitability for developing AR applications. Unity will be used for developing the application because it has the necessary features to support the development of AR applications with the help of AR Foundation. In addition, Unity will allow the development team to integrate features like animation, interactive design, and visual effects, which are necessary to develop interactive AR-based educational applications. When compared to other development environments like Unreal Engine, Unity will allow the development team to work efficiently with fewer documentation issues to focus on developing AR applications. In addition, Unity will allow the development team to efficiently design 3D assets with the help of Autodesk Maya and Blender to design plant models with animation features to optimize the development process. These software applications will allow the development team to efficiently design plant models with animation features to optimize the development process.

The process of data collection for the purpose of the present research would emphasize the way young learners interpret the visual cues related to the states of plants. The target group for the present research would include Sri Lankan international school students belonging to the 6-year-old category and following the Cambridge curriculum for Grade 1 Science, along with teachers' opinions related to the educational value of the proposed method of visualization. The process of data collection would involve observational testing sessions for the students using the AR storybook application. Feedback would be related to the ease of interpretation of the health conditions of the plants, the level of engagement of the children, and their level of understanding of the concepts of plant growth. Ethical considerations must be considered for the purpose of the present research, as the target group includes young learners. Therefore, the process of data collection would be conducted after taking the proper consent of the parents of the target group. No personal data would be included for the purpose of the present research.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution will be realized in the form of a handheld Augmented Reality (AR) interactive storybook system that aims to facilitate ecological learning for Sri Lankan international school children in grade one, where the children are six years old and are following the Cambridge curriculum for grade one science. The system will comprise multiple components that are integrated in such a way that they are able to simulate the behavior of plant growth, interaction with the environment, and emotionally interpretable visual feedback. The system architecture, as shown in Figure X, highlights the high-level system architecture that indicates the components of the proposed solution and the information flow between the input layer, processing components, and the output layer.

The architecture has three main layers: input layer, core processing layer, and output layer. The input layer consists of storybook pages, camera input, mobile sensor input, user interaction input, and learner progress information. Input information is processed by four main components of the system: Hybrid Interaction Framework, Affective Visualization Model, Learning Narrative Engine, and Context-Aware Simulation Engine. A Plant State Manager acts as a data layer to store information regarding the current biological state of the virtual plant. The current state of the virtual plant is updated by interaction events and environmental conditions. The Affective Visualization Model retrieves information from the Plant State Manager to produce relevant visual and audio cues for the child user. The processed information is then sent to the child user in the form of AR visual cues, audio cues, and learning cues. The modular architecture allows for easy interaction among system components and facilitates the integration of simulation, storytelling, and affective visualization in the AR learning environment.

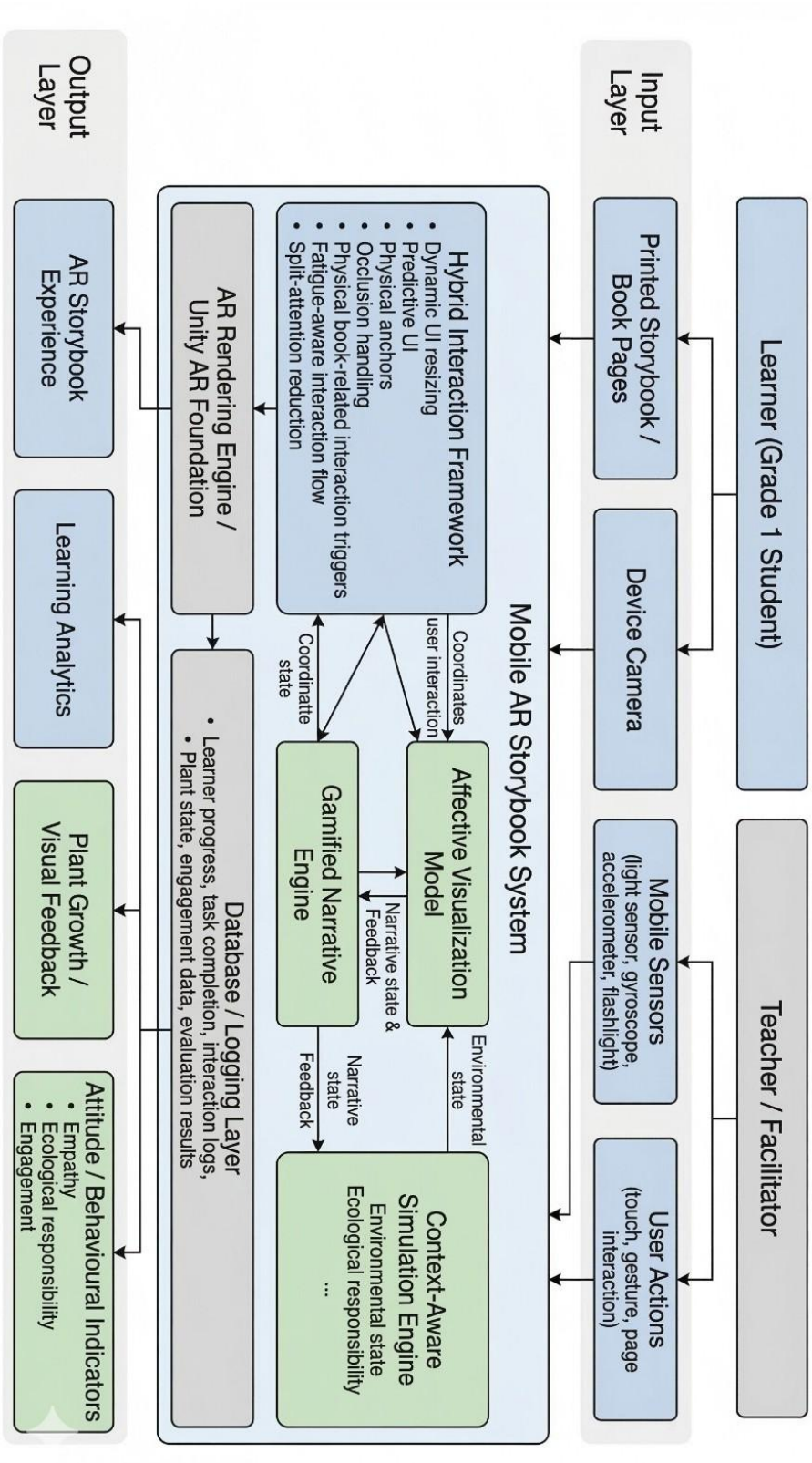


Figure 1: System Diagram of Components

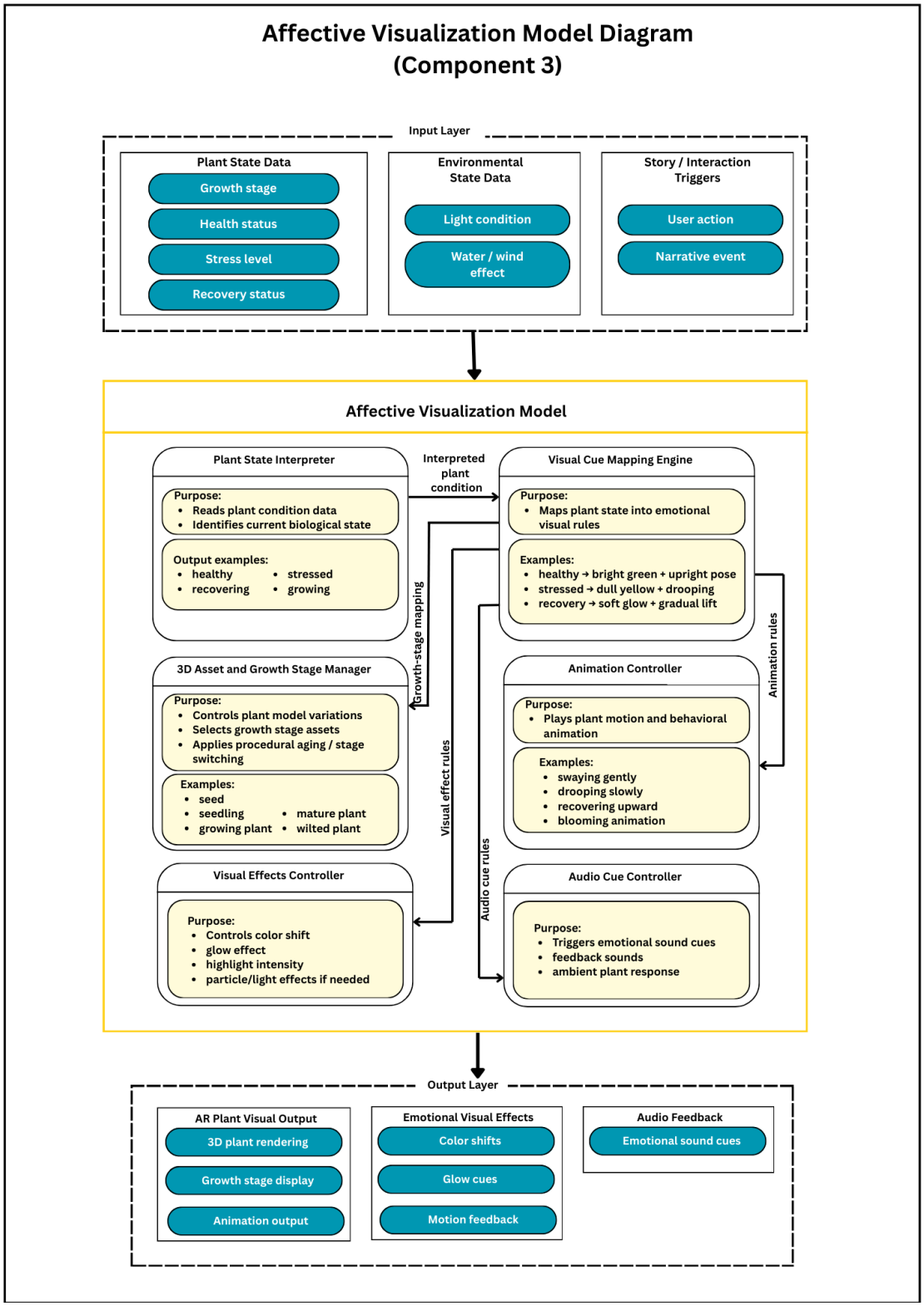


Figure 2 : High-level architecture of the affective visualization model

The Affective Visualization Model is an important part of the proposed AR learning framework and deals with the conversion of biological plant state information into emotionally interpretable visual cues. This module will be responsible for providing an interactive 3D model for representing plant growth stages, health conditions, and environmental responses. The proposed model will be able to convert plant state information such as health, stress, and recovery into visually interpretable cues such as color changes, glow, motion, and audio. The high-level architecture of the proposed Affective Visualization Model, as presented in Figure X, represents the internal modules responsible for interpreting plant state information, generating emotional visual cues, and providing AR-based visual and audio outputs.

As may be seen from the architecture, the Affective Visualization Model receives the plant state data, environmental parameters, and interaction triggers from the core system. These inputs from the core system are processed initially by the Plant State Interpreter, which interprets the biological state of the plant. Then, the interpreted plant state is processed by the Visual Cue Mapping Engine, which maps the appropriate visual and audio cues based on the defined visualization rules. Other modules that assist the Affective Visualization Model include the 3D asset and growth stage manager, animation controller, visual effects controller, and audio cues controller. These modules assist the Affective Visualization Model in generating the final visual representation of the plant in the context of augmented reality. As may be seen from the architecture, the outputs of the Affective Visualization Model include the animated 3D visuals of the plant, the color and glow effects that define the health conditions of the plant, and the audio cues that define the emotional feedback.

The proposed system architecture is intended to be modular and scalable to accommodate the addition of new learning modules, plant species, and educational content without significant changes to the system architecture. This is because the proposed system architecture splits the interaction, simulation, narrative, and visualization components of the system into different modules. This would enable the addition of new features, such as new plant growth scenarios, environmental simulation, or new chapters for the storybook, by extending the modules. This would enable the system to accommodate the addition of

new features for other science learning areas besides plant growth. From a security point of view, the proposed system architecture would minimize the storage of user personal data since the system would interact mainly through the mobile application. If the system stores the user learning progress, the data would be anonymized and used for educational evaluation. In addition, the proposed system would obtain parental and institutional approvals for user testing of the system for children, thus following the guidelines for ethical research.

5. USER REQUIREMENTS

5.1. Stakeholder identification

The proposed system's main users are Sri Lankan international school children who are six years old and are following the Cambridge Grade 1 Science curriculum. Children at this early age are introduced to basic concepts related to plant growth and environmental awareness. However, it has been found that conventional learning methods, such as referring to books, viewing images, and listening to teachers, make it difficult for young children to understand how plants grow and how they react to changes in the environment. The proposed AR interactive storybook system has been developed to solve this problem. The proposed AR interactive storybook system will allow children to understand how plants grow by using Augmented Reality technology to visualize the growth stages of plants, understand the health conditions of plants, and interact with the virtual plant objects placed in their environment. The proposed AR interactive storybook system will allow the child to understand plant conditions intuitively with the help of visual animations, colors, glow, and audio.

Teachers constitute the next important group of stakeholders that the proposed system is designed to impact. In the context of the primary school setting, the role of the teacher is significant as they facilitate the students' interaction with the learning activities, thereby ensuring that the educational goals are met. In the context of the proposed AR interactive storybook system, the role of the teachers is significant as they would be able to use the proposed system as a learning aid while conducting their science classes with respect to the growth of plants and the environment. With the proposed storytelling and interactive visualization features, the teachers would be able to explain the biological process to the students in a more interesting and comprehensible way. In addition, the teachers would be able to observe the interaction of the students with the AR content and use the visual feedback provided to the students to ensure that the concepts of the sciences are clearly imparted.

Besides the students and the teachers, there are other stakeholders who are involved in the successful deployment of the system. There are the schools and educational institutions

that are considered the implementation stakeholders, as they are responsible for integrating the AR interactive storybook into the classroom. There are also the parents who can be considered indirect stakeholders, as the system is geared towards creating environmental awareness among children. Moreover, the developers and researchers who are involved in the project are considered critical stakeholders, as they have the responsibility of developing the AR application, developing the affective visualization model, and improving the system based on user feedback and results from the evaluation. All these stakeholders are responsible for the successful adoption of the proposed system.

5.2. Functional requirements

The proposed AR interactive storybook system should be able to perform a set of functional requirements that allow for interaction between the learner and the learning environment. These requirements are the basic actions that the system should be able to perform to allow for an effective augmented reality learning experience for the children. The proposed system will be able to allow for interaction between children and plants by providing storybook interaction and a digital AR visualization. This will allow children to observe the growth of plants and changes in the environment. The proposed system will be able to use the camera on a mobile device to allow children to interact with the 3D content and observe the growth stages of plants, understand the effect of the environment on plants, and identify the conditions affecting plants. Each of these functional requirements will be geared towards ensuring that children are able to learn ecological concepts by interacting with the learning environment rather than simply observing. The following are the functional requirements for the proposed AR interactive storybook system.

i. Storybook Page Recognition

The system shall recognize physical storybook pages or AR markers using the mobile device's camera. When a learner scans the page of the storybook, the system shall recognize the marker or the image of the page of the storybook. Once the system recognizes the marker or the image of the page of the storybook, the augmented reality scene shall then be activated and matched to

the physical page of the storybook. This system shall then enable the integration of physical storytelling and augmented reality visualization.

ii. AR Plant Visualization

The system shall display interactive three-dimensional plant models within the real-world environment using augmented reality technology. When the system recognizes the storybook pages or the marker, it shall then display a 3D model of a plant. This 3D model of a plant shall then be displayed as if it is attached to the physical environment.

iii. Plant Growth Stage Representation

The system shall allow the visualization of the growth stages of the plant. This will enable the learners to understand the growth stages of the plant. The growth stages may include the seed stage, the seedling stage, the growing plant stage, and the mature plant stage. Each growth stage should be clearly identifiable based on the size, shape, and look of the plant.

iv. Plant Health Condition Visualization

The system shall allow the learners to understand the health condition of the plant. There may be conditions where the plant is growing healthily, the plant is under environmental stress, or the plant is recovering. All the conditions may be represented with the use of visual cues. For example, the plant may look erect and fresh if it is growing healthily.

v. User Interaction with AR Content

The system shall enable learners to interact with the augmented reality plant using simple gestures and device interactions. Learners shall be able to scan storybook pages, tap the augmented reality plant model, or move the device to interact and look at the plant from different angles. This promotes active

learning and enables children to learn about concepts of plant growth in a more engaging and participatory way.

vi. Narrative-Based Learning Guidance

The system shall enable learners to receive story-based guidance and feedback to help learners understand concepts of plant growth and the environment. The system shall enable the presentation of learners' instructions, hints, or explanations related to the care and growth of plants using the storybook narrative.

vii. Environmental Simulation

The system shall provide an interface that allows for the simulation of various environmental factors that affect plant growth and behavior. For instance, the application may use the movement of devices and lighting as an environmental factor that affects the behavior of the plant. This allows the learner to understand the relationship that exists between the plant and the environment.

viii. Emotional Visual Feedback

The system shall provide an interface that allows for emotional visual feedback, enabling the system to provide an intuitive interface for the learner to understand the health condition of the plant. This includes visual feedback such as color changes, glow, and animation of the plant, allowing young children to understand the condition of the plant without the need for scientific explanations.

5.3. Non-functional requirements

Non-functional requirements are a set of requirements that describe the quality attributes and standards for which the system should operate to be effective within a real-world educational scenario. Non-functional requirements are not related to what the system should or should not perform, as with functional requirements. Instead, non-functional requirements are related to the way the system should perform. Non-functional requirements for the proposed AR-based interactive storybook system are related to ensuring that the system should operate with high reliability, usability, security, and smooth augmented reality experiences on mobile devices for use within a classroom scenario. Since the target users for this system are six years old, usability and accessibility are critical non-functional requirements to be considered to ensure that the system's interface and visual feedback are simple, intuitive, and entertaining. Performance non-functional requirements are critical for AR-based applications since they need to render 3D graphics, animations, and other visual effects smoothly without any delays that might interfere with the learning process. Security and privacy are other non-functional requirements that need to be considered when recording learner interaction data to ensure that the system operates within acceptable ethics.

Table 2 : Non-functional requirements of the proposed system

Non-Functional Requirement	Description	Justification for Affective Visualization Component
Usability	The system must provide a simple and intuitive interface suitable for children aged six.	The Affective Visualization Model uses clear visual cues such as color changes, glow effects, and animations that are easily understood by young learners.
Performance	The AR application must render 3D models and animations smoothly on mobile devices without noticeable delays.	Efficient 3D asset management and optimized visual effects ensure responsive visualization of plant states.
Reliability	The system should consistently display correct plant states and visual feedback during interaction.	The visualization component relies on plant state data to accurately represent plant health conditions through animations and visual cues.
Scalability	The system should allow additional plant species, storybook chapters, or visual cues to be added in future updates.	The visualization module can be extended to support new plant models, growth stages, and additional emotional feedback cues.
Security	The system must protect user privacy and avoid collecting sensitive personal data.	The visualization component processes only plant state information and interaction events, without storing personal user data
Accessibility	The system should provide visual and audio cues that support different learning preferences.	The component combines animation, color changes, glow effects, and sound cues to improve comprehension and accessibility.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The primary target market for the proposed solution is international schools in Sri Lanka that follow the Cambridge Primary Science curriculum, especially schools that seek interactive and technology-supported learning tools for early primary education. The system is designed as a handheld AR storybook that runs on standard ARCore- and ARKit-enabled smartphones and tablets, which makes it suitable for schools that already use mobile devices or tablets in classroom activities. Although the AR storybook is developed according to the Cambridge Grade 1 Science syllabus in international schools, it is also appropriate for 6-year-old children who enjoy learning about plants, because the concept, visuals, and interaction style are intended for early learners.

Table 3 : Customer Persona: School Decision-Maker

Category	Details
Persona Name	School Decision-Maker
Role	Principal / Sectional Head / ICT Coordinator / Primary Curriculum Coordinator
Type	Primary Customer
Profile	Responsible for selecting and approving educational technology solutions for the school.
Needs	A reliable, curriculum-aligned, safe, and innovative learning tool that improves student engagement and supports classroom teaching.
Goals	To introduce effective digital learning solutions that enhance educational quality and support modern teaching methods.
Pain Points	Difficulty finding age-appropriate AR tools that are educationally meaningful, easy to implement, and suitable for young learners.
How the System Helps	The AR storybook provides a research-based, child-friendly, and curriculum-relevant learning solution for Grade 1 science education.

Table 4 : Customer Persona: Grade 1 Science Teacher

Category	Details
Persona Name	Grade 1 Science Teacher
Role	Early Primary Science Teacher
Type	Secondary Customer / Direct User
Profile	Uses the system in classroom teaching to help students understand plant-related science concepts in an interactive way.
Needs	A classroom-friendly tool that is easy to use, engaging for children, and supportive of concept understanding.
Goals	To make science learning more interesting, understandable, and memorable for Grade 1 learners.
Pain Points	Traditional teaching methods may not always keep young learners engaged or help them clearly visualize plant-related concepts.
How the System Helps	The AR storybook combines storytelling, visuals, and guided interaction to make science lessons more engaging and easier to understand.

Table 5 : Customer Persona: Parent of a Grade 1 Learner

Category	Details
Persona Name	Parent of a Grade 1 Learner
Role	Parent / Guardian
Type	Indirect Customer
Profile	Supports the child's learning at home and values meaningful educational experiences beyond the classroom.
Needs	A safe, engaging, and educational tool that helps the child learn effectively in a guided or supervised setting.
Goals	To support the child's curiosity, enjoyment, and understanding of science concepts at home.
Pain Points	Limited access to interactive educational tools that are both enjoyable and academically useful for young children.
How the System Helps	The AR storybook supports emotional engagement, guided learning, and progress-related feedback, making it useful for supervised home learning as well.

6.2. Value Proposition

The main value proposition of the proposed system is that it offers a child-centered, safe, and engaging AR learning experience that combines physical storybooks with interactive digital content to improve understanding of plant-related science concepts. The proposed solution already emphasizes that printed storybook pages, camera input, mobile sensor data, and guided interactions are integrated to create an AR learning experience that is more meaningful than static textbook diagrams. In addition, the system is differentiated by its coordinated research modules, particularly the Hybrid Interaction Framework, which addresses touch accessibility, anchoring between physical pages and 3D content, split-attention reduction, occlusion handling, and fatigue-aware interaction support.

From a commercialization perspective, this gives the product clear value for schools: it is not just an AR storybook, but a research-driven educational tool designed specifically for

young learners' usability, comfort, and engagement needs. This makes it more attractive than generic AR applications that may focus mainly on visual novelty without considering children's fine motor skills, physical comfort, or cognitive load.

6.3. Revenue Model

A suitable revenue model for this product is a business-to-school licensing model. Under this model, schools would pay an annual or term-based subscription for access to the AR storybook application, learning content, teacher support features, and software updates. This approach is practical because the system is intended for structured educational use rather than one-time casual entertainment. A second possible model is a per-classroom or per-device licensing structure, where schools pay based on the number of student devices or class groups using the system.

An additional revenue stream could come from content expansion packs, such as new storybooks, additional science topics, or curriculum-aligned activity sets for later grades. Since the proposed architecture is modular and includes multiple coordinated research components, the solution has the potential to be extended beyond the initial plant-growth topic into broader environmental or science learning domains. In the longer term, the product could also be offered through partnerships with educational publishers or private tuition and learning centers.

6.4. Cost Estimation

The main cost categories of commercialization would include software development and refinement, content production, 3D asset and animation development, testing and quality assurance, teacher onboarding materials, and maintenance/support costs. The project requires practical development with Unity 3D (2024 LTS), AR Foundation, ARCore, and ARKit, as well as ongoing prototyping, iteration, and testing. These factors suggest that the early commercialization stage would involve costs related to continued technical development, optimization, and device compatibility testing.

There would also be costs for producing high-quality storybook pages, visual assets, and classroom-ready deployment materials. If the product is marketed to schools, teacher guides, set up instructions, and user support documentation would also be necessary. In the pilot stage, costs may be relatively high per user because the product is still being refined. However, once the core application and content pipeline are established, the marginal cost of distributing the software to additional schools would be lower than that of physical-only learning tools.

6.5. Pricing Strategy

The pricing strategy should be based on institutional affordability and educational value. A practical approach would be to offer a school subscription package with different tiers depending on the number of users, classes, or content modules required. Because the target customers are schools rather than individual mass-market consumers, the pricing should remain reasonable enough for adoption while also reflecting the research-based and curriculum-aligned value of the product. A pilot pricing strategy may also be used in the initial stage, where selected schools receive the system at a discounted rate in exchange for implementation feedback and case-study data. This would support both early adoption and product refinement.

6.6. Competitive Advantage

The proposed solution has several competitive advantages. First, it combines physical storytelling with handheld AR in a way that is explicitly designed for young learners rather than adapted from adult-centered AR systems. The Hybrid Interaction Framework includes predictive UI, dynamic button resizing, physical book-related interaction triggers, and fatigue-related interaction design to improve usability and comfort for children in handheld AR learning. This gives the product a strong advantage over many existing AR learning tools that focus mainly on visual interaction without addressing motor accessibility, ergonomic comfort, or split-attention reduction.

Second, the product is built as an integrated educational ecosystem rather than as a standalone AR animation tool. The complete system includes a Hybrid Interaction

Framework, an Affective Visualization Model, a Narrative Engine, and a Context Aware Simulation Engine working together as coordinated modules. This means the product can offer emotionally responsive visuals, story-driven engagement, and real-world sensor-based interaction in addition to usability improvements. Third, because the solution is intended for early science education and is aligned to a school curriculum context, it has stronger classroom relevance than general-purpose AR entertainment applications.

6.7. Intellectual Property Considerations

The intellectual property (IP) of the proposed product would likely include several protectable elements: software implementation, the interaction design logic, the storybook content, the 3D assets and animations, and possibly the novel integration of hybrid interaction mechanisms such as adaptive UI resizing, physical anchors, book related triggers, and fatigue-aware flow design. These aspects may support future claims for software ownership, design protection, or publication-based research recognition. At the commercialization stage, it will be important to clearly determine the ownership of the software, assets, and content among the student researchers, the university, and any future industry partners. Copyright protection would apply to the storybook design, visuals, animations, and source code, while trademark protection may later be considered for the product name and brand identity. If the final implementation contains sufficiently novel technical methods, there may also be potential for patent-related discussion, although that would require legal and institutional review. In addition, care must be taken to avoid infringement when using third-party libraries, asset packs, or audio resources during product development.

7. BUDGET AND JUSTIFICATION

7.1. Hardware Costs

The development and testing of the AR interactive storybook system need hardware devices that are appropriate for the development and testing of augmented reality-based applications. Since the system will be implemented on mobile devices, a compatible Android smartphone or tablet with ARCore will be required for the development and testing of the AR storybook. Other hardware requirements include basic hardware devices used for development and testing activities. These hardware devices will enable the developers to test the AR tracking, interaction, and visualization elements to ensure the correct functionality of the AR storybook.

Table 6 : Hardware Cost Estimation

Item	Quantity	Unit Cost (Rs.)	Justification
AR-supported smart device (testing device)	1	35,000	Required to test AR application performance and compatibility on a mobile device used by students.
Phone tripod / stand	1	3,000	Helps stabilize the device during AR marker testing and recording demonstrations.
Charging cables / accessories	1	500	Required for continuous device usage during development and testing.
Subtotal Software Cost: Rs. 38,500			

7.2. Software Licenses

To create the application, the following software tools will be required. These tools will be used to create the application, design the application, and create the content. Most of the tools used in this application allow the developers to use free versions of the software tools. These free versions of the software will help the developers save money. To create the application, the Unity platform will be used. To create the content used in the application, Blender and Maya software tools will be used. These tools will be used to create the 3D plant models that will be used in the application. Programming and debugging the application will be done with the Visual Studio or Visual Studio Code software tools.

Table 7 : Software License Cost Estimation

Software	License Type	Cost (Rs.)	Justification
Unity Personal License	Free	0	Core platform used for developing the AR application and integrating all system components.
Blender	Free	0	Used for creating and animating 3D plant models for visualization.
Autodesk Maya (Student License)	Free	0	Used for advanced modeling and animation of plant assets.
Visual Studio / VS Code	Free	0	Used for programming and debugging application scripts.
Subtotal Software Cost: Rs. 0			

7.3. Cloud Services

The cloud services may be employed for optional backend features such as application update, data storage, and content delivery. Even though the primary AR functionality is performed locally on the device, cloud services may be employed for the distribution of the content and user feedback storage in the testing phases. These cloud services may ensure that system updates, additional modules of the story, and educational content may be distributed efficiently. Employing basic cloud infrastructure may ensure that the system is scalable, and operational costs remain low.

Table 8 : Cloud Services Cost Estimation

Software	License Type	Cost (Rs.)	Justification
Cloud storage (Google Cloud / AWS basic tier)	6 months	3,000	Used for storing prototype data, updates, and testing resources.
Domain / hosting for project demo	1 year	0	Allow hosting a project demo page or documentation for demonstration and evaluation.
Subtotal Cloud Services Cost: Rs. 3,000			

7.4. Human Effort Cost

Human effort refers to the time and labor spent in developing, testing, and perfecting the AR interactive storybook system. It includes developing the software, designing 3D assets, animation, and testing the system. As the developers are students, the labor costs are based on the reasonable values that could be spent on developing the system, not the actual costs that could be incurred by professionals. There could be some traveling costs involved in going to schools to observe, test, and demonstrate the system, but they are minimal.

Table 9 : Human Effort Cost Estimation

Task	Estimated Cost (Rs.)	Justification
Software development and integration	4,500	Time spent developing AR application features and integrating system modules.
3D modeling and animation development	3,000	Creation of plant models, animations, and visual assets for the AR visualization component.
System testing and debugging	2,000	Time required for testing system functionality and fixing technical issues.
Travel expenses for school visits / testing	3,000	Transportation costs for visiting schools during observation or testing phases.
Subtotal Human Effort Cost: Rs. 12,500		

Table 10 : Total Estimated Budget

Estimated Total Budget	
Category	Cost (Rs.)
Hardware Costs	38,500
Software Licenses	0
Cloud Services	3,000
Human Effort Cost	12,500
Total Estimated Budget: Rs. 54,000	

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) of the proposed research is organized according to the four integrated components of the handheld AR storybook system. In the proposed solution, the system operates through printed storybook pages, camera and sensor inputs, and user actions, while the process layer is divided into four coordinated modules: the Hybrid Interaction Framework, the Affective Visualization Model, the Gamified Narrative Engine, and the Context-Aware Simulation Engine. The final outputs are expected to include the AR storybook experience, learning analytics, and attitude-related indicators. This modular structure provides a clear basis for dividing work among the team members and supports systematic development and integration of the overall product.

Table 11 : Work Breakdown by Research Component

Member	Component	Task Breakdown
IT22641410 - Wijayarathne K.N.	Hybrid Interaction Framework	Design and implement the dynamic UI mechanism for children's fine motor support; develop physical anchors and occlusion handling for printed storybook interaction; design fatigue-aware interaction flow to reduce prolonged arm strain; conduct usability testing related to cognitive load, engagement, and split attention reduction
IT22179258- Palliyaguruge S.N.	Affective Visualization Model	Design AR-optimized 3D plant assets; create animations for plant health, stress, and recovery; apply visual and audio effects to improve emotional readability; conduct child-centered AR trials to refine interpretation and empathy-oriented visual performance.

IT22237804-Amarawardhane A.S.	Gamified Narrative Engine	Develop the branching story structure; create AI/rule-based dialogue and feedback according to plant condition and context; implement time-delta persistence for plant growth and recovery; design redemption-based rewards and evaluate environmental attitudes and responsibility.
IT22626974-Dissanayaka D.M.D.D.	Context-Aware Simulation Engine	Develop light-sensing functionality; create algorithms to simulate plant growth based on environmental conditions and user actions; optimize sensor aggregation for efficiency; implement flashlight-assisted growth support and gyroscope-based wind/water simulation.
All Members	Integration and Evaluation	Integrate the four modules into one handheld AR storybook system; perform joint system testing, debugging, user evaluation, refinement, presentation preparation, and final product completion

Table 12 : Balanced Workload Distribution

Work Area	Workload Distribution Explanation
Component Ownership	Each member is assigned one major research component, which creates clear responsibility and avoids unnecessary overlap. This supports balanced distribution across interaction, visualization, narrative, and simulation development.
Technical Development	The work is balanced because all four members develop a comparable subsystem with its own design, implementation, and testing responsibilities.
Integration Work	Integration is shared because the final system depends on interaction among all four modules inside one AR storybook platform. This includes combining sensor input, page tracking, story flow, visualization, and usability support.
Testing and Evaluation	Evaluation is also shared, since the project collects observation data, interaction logs, plant growth status, and pre/post outcomes, and requires ethical handling of child participants.
Final Deliverables	Report writing, presentation preparation, website completion, and final refinement are group responsibilities, ensuring that no single member carries all final-stage work.

9. GANTT CHART

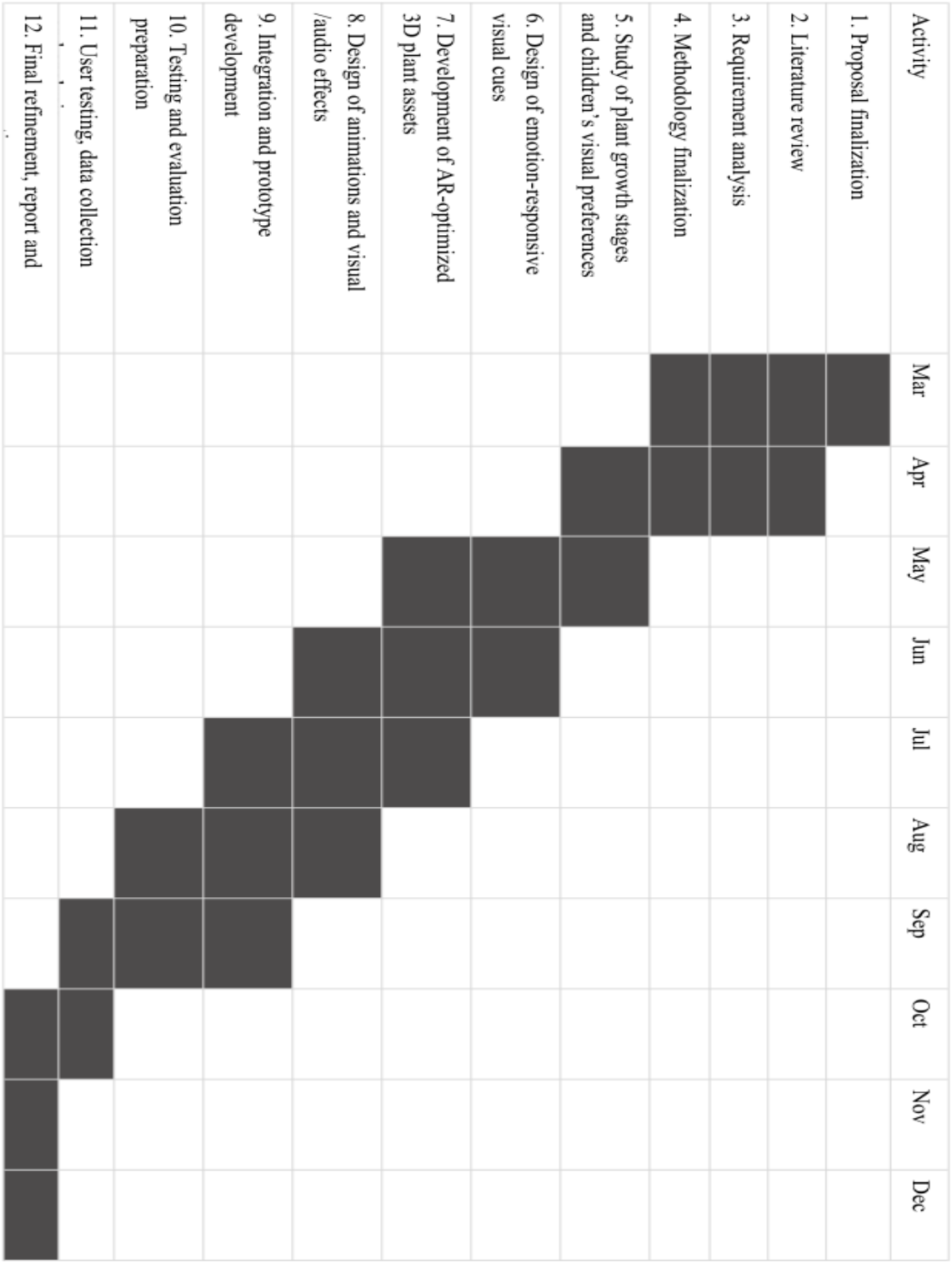


Figure 3 : Gantt Chart (March 2026 – December 2026)

The Gantt chart presents the planned schedule of activities for the research project from March to December 2026. It provides a clear timeline for the major phases of the study, including proposal finalization, literature review, requirement analysis, design and development of the affective visualization model, testing, data collection, analysis, and final documentation. The chart is structured to ensure that each stage of the research is completed within the given academic timeframe while maintaining a logical sequence of tasks. It also helps monitor progress, manage deadlines effectively, and ensure that the project remains aligned with key academic milestones and expected research outcomes.

10. REFERENCES LIST

[1] UNESCO, *Education for Sustainable Development: A Roadmap*, Paris, France: United Nations Educational, Scientific and Cultural Organization, 2020.

[2] C. Ibáñez and C. Delgado-Kloos, "Augmented reality for STEM learning: A systematic review," *Computers & Education*, vol. 123, pp. 109–123, 2018.

[3] M. Paredes-Velasco, J. A. Velázquez-Iturbide, and M. Gómez-Ríos, "Augmented reality with algorithm animation and their effect on students' emotions," *Multimedia Tools and Applications*, vol. 81, pp. 30589–30612, 2022.

[4] United Nations, "Transforming our world: The 2030 Agenda for Sustainable Development," United Nations, 2015. [Online]. Available: https://sdgs.un.org/goals

[5] H. Y. Chang, H. C. Wu, and Y. S. Hsu, "Trends and research issues of augmented reality studies in science education: A review of academic journal publications," *Computers & Education*, vol. 123, pp. 1–16, 2018.

[6] G. Lampropoulos, K. Keramopoulos, and K. Diamantaras, "Enhancing STEM education through augmented reality: A systematic literature review," *Education and Information Technologies*, vol. 27, no. 2, pp. 1–23, 2022.

[7] T. I. Ladykova, "Augmented reality in environmental education: A systematic review," *EURASIA Journal of Mathematics, Science and Technology Education*, vol. 20, no. 3, 2024.

[8] E. Öztürk, “Can environmental education supported by augmented reality affect students’ environmental knowledge, attitudes and behaviors?” *European Proceedings of Educational Sciences*, vol. 15, pp. 152–160, 2023.

[9] J. H. Lo, C. Y. Lai, and H. Y. Hsu, “The study of AR-based learning for natural science inquiry activities in elementary school from the perspective of sustainable development,” *Sustainability*, vol. 13, no. 11, p. 6283, 2021.

[10] Z. J. Jhuang, Y. C. Lin, and Y. T. Lin, “Effects of developing an interactive augmented reality plant structure experiment system for elementary natural science courses,” *International Journal of Information and Education Technology*, vol. 14, no. 8, pp. 1050–1057, 2024.

[11] Y. Wen, X. Chen, and J. Chen, “Integrating augmented reality into inquiry-based learning: Effects on learning achievement, motivation and cognitive load,” *Education and Information Technologies*, vol. 28, pp. 1–18, 2023.

[12] N. Aziz, S. Z. Ahmad, S. Binsaleh, and W. W. R. Z. Wan A. Rahman, “Heuristic evaluation on affective 4-dimensional augmented reality mathematics for children with low vision,” *International Journal of Information and Education Technology*, vol. 13, no. 2, pp. 1–7, 2023.

[13] J. L. Plass, R. Heidig, B. D. Homer, and E. O. Hayward, “Emotional design in multimedia learning: Effects of shape and color on affect and learning,” *Learning and Instruction*, vol. 29, pp. 128–140, 2014.

11. APPENDICES

The appendices of this proposal contain materials that provide supplementary information to the body of the report. These materials provide additional information that pertains to the research design, system structure, evaluation, and ethical considerations. These materials are included in the proposal to provide better clarity, completeness, and feasibility of the research design. Considering that the research design pertains to the development and evaluation of a handheld AR storybook system for kids, the appendices of the proposal have an important function in providing information that pertains to the research design, system structure, and procedural considerations that are part of the research process.

The appendix section of the proposal is divided into five sections, namely, survey instruments, diagrams, preliminary UI mockups, risk analysis, and ethical considerations.

11.1. Survey Instruments

The survey tools included in the appendices would be utilized to aid the evaluation process of the proposed research. These survey tools would be utilized to gather data concerning the children's comprehension of the concepts of plant growth, interpretation of affective visual cues, interaction with the AR environment, and emotional responses to the visualization model. Apart from the survey tools designed for the children, other survey tools could also be developed to obtain data from teachers or observers concerning the usability concerns, learning behaviors, and interactional qualities during the testing sessions. As discussed, the project plan already indicates the utilization of pre-test and post-test questionnaires, as well as the utilization of observation-based data collection and system logs, as part of the evaluation process.

The survey instruments will be designed using age-appropriate language, short response formats, and simple rating methods so that children in the target age group can respond comfortably. Where necessary, visual scales such as smiley-face ratings or simplified multiple-choice formats may be used to improve response accuracy and reduce confusion. These materials will help measure whether the visualization cues successfully

communicate plant health, stress, and recovery states in a way that is understandable to young learners. The appendix may include draft versions of the following instruments

Table 13 : Survey Instruments for Research Evaluation

Instrument	**Purpose**	**Target Respondent**
Pre-test questionnaire	Measure prior knowledge of plant growth and environmental concepts	Child participants
Post-test questionnaire	Measure learning improvement after interacting with the AR system	Child participants
Engagement survey	Assess interest, enjoyment, and emotional connection with the plant character	Child participants
Observation checklist	Record behavioral responses, hesitation, excitement, confusion, and interaction patterns	Researcher / observer
Teacher feedback form	Collect views on educational suitability and classroom relevance	Teacher / academic reviewer

11.2. Detailed Diagrams

This subsection will include the technical and conceptual diagrams that will give a clearer view or representation of the proposed system aside from what has been described in the main chapters. These diagrams are quite important since the proposed system includes a number of modules that work together to produce the expected output. These modules include the Hybrid Interaction Framework, Affective Visualization Model, Gamified Narrative Engine, and Context-Aware Simulation Engine. The uploaded project material already shows a layered structure with input, process, and output layers, as well as an internal structure for the Affective Visualization Model.

The diagrams to be included in the appendices may include the overall system architecture, data flow diagrams, module interaction diagrams, activity flow diagrams, and the internal high-level architecture for the Affective Visualization Model. With respect to your individual component, the most important diagrams will probably include those showing the interpretation and translation of plant state data into color, glow, animation, motion, and audio. These diagrams will help the reviewer understand the structure of the proposed system and the role of your component within the whole AR learning framework.

The appendix may include diagrams such as:

- Overall system architecture diagram
- High-level architecture of the Affective Visualization Model
- Data flow diagram for plant state interpretation
- Sequence diagram for user interaction and visual response
- Activity diagram for child interaction with the AR storybook
- Integration diagram showing communication among all four modules

11.3. Preliminary UI Mockups

The preliminary UI mockups included in the appendices will be used to describe the proposed design for the mobile application. In addition, since this study will be child-centered with an emphasis on education, it will be important to have an interface that not only looks good but also captures the emotions of the child. It will also be important to have an interface that can be easily navigated by the child. The description has already emphasized the importance of age considerations, interactive cues, and the use of storybooks in learning. Therefore, it will be important to include the mockups to describe the proposed design for the application.

For my component, it will be important to include mockups that describe how the healthy, stressed, and recovering states of the plant are visually represented. It can include differences in plant color, glow intensity, facial cues if used, plant movement style, and sound feedback. The purpose of including the mockups is not to describe how the user

interface will look, but to describe the proposed user experience and how emotional readability will be supported.

A suitable list of mockups to include is shown below:

Table 14 : Preliminary UI Mockups list of the Proposed System

Mockup	Description
Home screen	Entry point to the AR storybook application
Storybook page scanning screen	Interface used to detect and activate AR content
AR plant visualization screen	Main interaction screen displaying the animated plant
Plant condition feedback screen	Visual response view for health, stress, or recovery states
Progress or achievement screen	Child progress, rewards, or learning feedback
Settings / guidance screen	Parent, teacher, or child support instructions

11.4. Risk Analysis

The appendix for the risk analysis will outline the main challenges that may arise in order to ensure the successful completion of the research and development of this suggested system. This section will be included to prove that a realistic project plan has been prepared and that all possible problems are being thought about in advance. This project will involve AR interaction, affective visualization, child-centric testing, mobile performance constraints, and educational content development; thus, it will encompass both technological and research risks. The project documents already emphasize child usability, child cognition, devices, and ethics as important factors.

Table 15 : Risk Analysis of the Proposed Research

Risk	**Possible Impact**	**Mitigation Strategy**
Low-performance mobile devices	Lag, frame drops, or poor AR experience	Optimize 3D assets, reduce heavy effects, test on mid-range devices
Difficulty in interpreting visual cues	Children may misunderstand plant states	Refine color, animation, and audio cues through pilot testing
Tracking instability in AR	Misalignment of virtual content with storybook pages	Improve marker quality and test under different lighting conditions
Limited participant access	Delays in user evaluation with children	Coordinate early with schools, teachers, and guardians
Ethical approval delays	Postponement of testing activities	Prepare consent documents and approval materials in advance
Excessive cognitive load	Reduced usability and learning effectiveness	Use simple interactions, guided prompts, and age-appropriate interface design
Integration issues among modules	Delays in prototype completion	Conduct periodic integration testing across team components
Time management problems	Missed deadlines for presentations or reports	Follow the Gantt chart and monitor milestone completion regularly

11.5. Ethical Documents

This subsection will contain the ethical materials required to support responsible research involving children. Since the proposed study includes observation, questionnaire-based evaluation, and interaction testing with child participants, ethical safeguards are essential. The uploaded topic assessment form clearly states that the project requires ethical clearance, parental consent, child assent, anonymization of data, and compliance with institutional ethical requirements.

The ethical appendix will thus include the documents required to ensure the voluntary participation of the subject, the safety of the subject, and the proper explanation of the study to the relevant parties. The documents may include the participant's information sheet, parental consent form, child's assent form, privacy statement, and the approval of the ethics committee. The language used should be clear and understandable, especially if the documents are intended for children or parents. The ethical appendix will thus prove the intention of the proposed study to not only develop an effective educational AR tool but also to conduct the study with the required ethical standards.

The following ethical documents may be included:

- Parent or guardian consent form
- Child assent form
- Participant information sheet
- Research purpose and procedure summary
- Data privacy and anonymization statement
- Institutional ethics review or approval documents
- School permission letter, where applicable